

# FENRIR ARENA

## Project Requirements Document

Fenrir Project | Online Gaming Website

| Project Type          | Total Files     | Backend          |
|-----------------------|-----------------|------------------|
| Browser Gaming Portal | 17 source files | Node.js + SQLite |

## 1. Project Overview

You are hired to build **Fenrir Arena**, a browser-based gaming portal for the Fenrir Project. Visitors can browse a catalog of casual games, play them directly in the browser, create an account, and compete on leaderboards. The site consists of 8 pages and 6 playable games, with a lightweight backend for accounts, scores, and leaderboards. All deliverables are source code files and documentation. No physical production is required.

| Field             | Value                                  |
|-------------------|----------------------------------------|
| Product name      | Fenrir Arena                           |
| Tagline           | Play. Compete. Conquer. (use verbatim) |
| Total pages       | 8                                      |
| Total games       | 6                                      |
| Backend framework | Node.js with Express                   |
| Database          | SQLite                                 |

### 1.1 Assets Provided by Client

| Asset               | Description                                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| reference_image.png | Sample Home Page layout - use as a visual guide for the header, hero banner, tagline placement, and featured games section |

## 2. Brand Guidelines

### 2.1 Colour Palette

| Use                | Colour Name    | Hex Code |
|--------------------|----------------|----------|
| Primary background | Charcoal Black | #1A1A1F  |
| Surface / header   | Wolf Gray      | #2E2E38  |
| Accent / buttons   | Ember Orange   | #FF7A30  |
| Secondary accent   | Frost Blue     | #4FB6E8  |
| Text               | Off White      | #F5F5F7  |

### 2.2 Mandatory Rules

- The site name "Fenrir Arena" must appear in the website header on every page
- The tagline "Play. Compete. Conquer." must appear verbatim on the Home Page and the About Page - do not rephrase
- The Home Page featured games section must show exactly 4 games
- The Game Library Page must show all 6 games with category filters: Arcade, Puzzle, Racing
- The layout must be usable at screen widths from 360px to 1920px

### 3. Pages (8 total)

Each page must be a separate file and must include the website header showing "Fenrir Arena". File format: HTML + CSS + JavaScript.

| # | Page            | File Name    | Required Elements                                                                            |
|---|-----------------|--------------|----------------------------------------------------------------------------------------------|
| 1 | Home            | index.html   | Hero banner, tagline, 4 featured games, Browse All Games button, follows reference_image.png |
| 2 | Game Library    | library.html | Grid of 6 games, filters (Arcade/Puzzle/Racing), game cards                                  |
| 3 | Individual Game | game.html    | Canvas, title, description, How to Play, leaderboard (top 10)                                |
| 4 | Login           | login.html   | Email field, password field, Login button, Sign Up link                                      |
| 5 | Sign Up         | signup.html  | Username, email, password fields, Create Account button, Login link                          |
| 6 | User Profile    | profile.html | Username display, best score table per game                                                  |
| 7 | About           | about.html   | Description text, tagline verbatim                                                           |
| 8 | Contact         | contact.html | Name, email, message fields, Send Message button                                             |

### 4. Games (6 total)

Each game must run inside an HTML5 Canvas element, must be controllable using keyboard or mouse input as specified below, and must display the player's current score during play. Each game is embedded on its own Individual Game Page.

| Game            | Category | Controls                              |
|-----------------|----------|---------------------------------------|
| Brick Breaker   | Arcade   | Left/Right arrow keys move the paddle |
| Snake Classic   | Arcade   | Arrow keys change direction           |
| Memory Match    | Puzzle   | Mouse click flips cards               |
| 2048 Tile Merge | Puzzle   | Arrow keys merge tiles                |
| Endless Runner  | Racing   | Spacebar triggers jump                |
| Sky Dash        | Racing   | Arrow keys steer, spacebar boosts     |

Score Submission: when a game ends, the final score must be sent to the backend save score endpoint. The score is saved only if the player is logged in.

## 5. Backend (Node.js + SQLite)

### 5.1 Endpoints

| Endpoint | Function                                                             |
|----------|----------------------------------------------------------------------|
| Sign up  | Accepts username, email, password; stores password as a hashed value |

|             |                                                                                   |
|-------------|-----------------------------------------------------------------------------------|
| Login       | Accepts email and password; returns a session that persists while the tab is open |
| Save score  | Accepts username, game name, score value, and date; stores the record             |
| Leaderboard | Accepts a game name; returns the top 10 scores sorted highest to lowest           |

## 5.2 Data Fields

| Record       | Fields                                 |
|--------------|----------------------------------------|
| User account | username, email, hashed password       |
| Score record | username, game name, score value, date |

## 6. File Naming Convention

Page files: [pagename].html / .css / .js (e.g. index.html, library.html, game.html, login.html, signup.html, profile.html, about.html, contact.html). Game files: games/[gamenam].js (e.g. games/brick-breaker.js, games/snake-classic.js, games/memory-match.js, games/2048-tile-merge.js, games/endless-runner.js, games/sky-dash.js).

## 7. Folder Structure

```
fenrir-arena/
  index.html, library.html, game.html, login.html, signup.html, profile.html, about.html, contact.html
  css/styles.css
  games/ (brick-breaker.js, snake-classic.js, memory-match.js, 2048-tile-merge.js, endless-runner.js, sky-dash.js)
  backend/server.js, backend/routes/accounts.js, backend/routes/scores.js, backend/database.sqlite
  README.md
```

## 8. Responsive Design Requirements

The layout must be usable on screen widths from 360px (mobile) to 1920px (desktop). All 8 pages must apply the colour palette in Section 2.1. All pages must include the website header showing "Fenrir Arena".

## 9. Final Deliverables Checklist

| Category | Description                                                                                    | Files |
|----------|------------------------------------------------------------------------------------------------|-------|
| Pages    | 8 pages: Home, Library, Game, Login, Sign Up, Profile, About, Contact                          | 8     |
| Games    | 6 games: Brick Breaker, Snake Classic, Memory Match, 2048 Tile Merge, Endless Runner, Sky Dash | 6     |
| Backend  | Accounts route + Scores route                                                                  | 2     |
| Docs     | README.md with local run instructions                                                          | 1     |

Total source files: 17 (8 pages + 6 games + 2 backend routes + 1 README)